

Solution Requirements

Date	12 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can access the crop recommendation system through the web application.	Medium
2	Authorization levels	Single-user access (Farmer/User) with no separate admin roles.	Low

3	External interfaces	Uses Flask web interface and a trained Machine Learning model for predictions.	High
4	Transactions processing	Accepts soil and weather parameters and processes them to generate crop recommendations.	High
5	Reporting	Displays recommended crop, prediction confidence, and crop information.	High
6	Business rules, etc.	Recommendation is generated only after all required input values are provided.	High
7	Compliance to laws or regulations	Follows standard Machine Learning and software development practices.	Medium
8	<i>Other</i>	User-friendly interface for quick crop prediction.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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1	Performance & Speed	Generate crop prediction quickly.	Response time < 2 seconds
2	Scalability	Support multiple users simultaneously.	Up to 100 concurrent users
3	Security & Data Privacy	User inputs are processed securely without storing personal data.	Secure data handling
4	Reliability & Availability	System should be available whenever users access it.	99% uptime
5	Usability & Accessibility	Simple and responsive interface for farmers.	Easy navigation and responsive design
6	<i>Other</i>	Maintainable and modular code structure.	Easy future enhancements